Reg No.:_______________ Name:_______________________

080EET201122006

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

Course Code: EET201

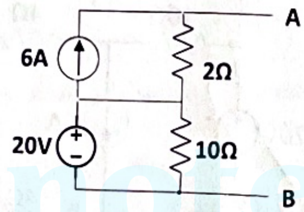
Course Name: CIRCUITS AND NETWORKS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1. State and explain maximum power transfer theorem in DC circuits. (3)
2. Replace the network given below with a single current source and a resistor. (3)



3. Explain the classification of series RLC circuits based on damping ratio. (3)
4. Obtain the expression for the voltage across a capacitor discharging through a resistor of resistance R. Assume that the initial voltage of the capacitor is V_0. (3)
5. Determine the voltage $v(t)$ across a 2Ω resistor, if the current is given by,. (3)

$$I(s) = \frac{2s+4}{s^2+4s+3}$$

6. Derive the s-domain equivalent circuit of a capacitor having an initial voltage of V_0. (3)
7. Explain the phenomenon of neutral shift in three phase 3 - wire systems. (3)
8. Derive an expression for the Q- factor of series resonant circuits. (3)
9. Express ABCD parameters in terms of Z parameters. (3)
10. Determine whether the two port network represented by the following network equations is reciprocal. (3)

$$V_1 = 3V_2 - 2I_2$$
$$I_1 = 4V_2 - 3I_2$$

PART B

Answer any one full question from each module. Each question carries 14 marks

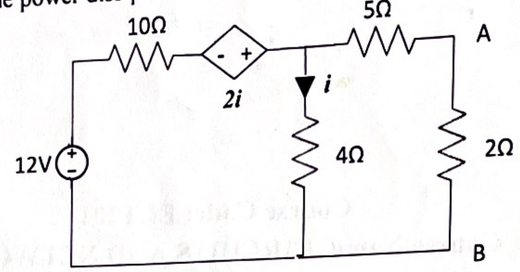
Module 1

11 For the network given below,

a) Obtain the Thevenin's equivalent circuit across the terminals A and B. (10)

(4)

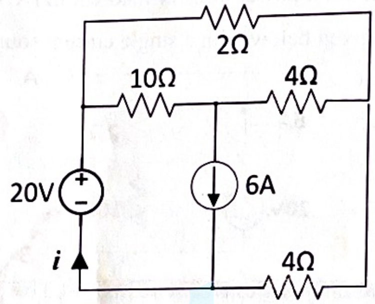
b) Determine the power dissipated in the 2Ω resistance.



12 In the circuit given below,

a) Find the current i using superposition theorem. (10)

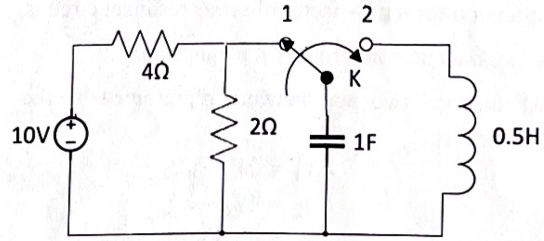
b) Determine the power supplied by the 20V source (4)



Module 2

13 a) A series RL circuit with R = 10Ω is connected to a 50V DC supply at $t = 0$. Determine the value of the inductance L if the current through the inductor attains 50% of its steady state value in 1 seconds. (7)

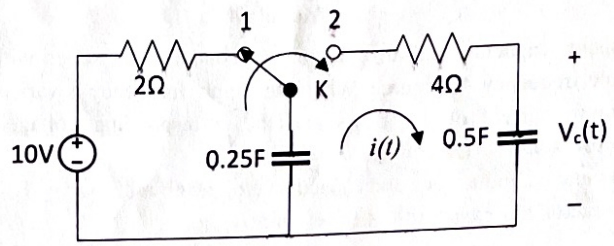
b) The switch K in the circuit given below has been at position 1 for a long time. At t = 0, the switch is moved to position 2. Determine the current flowing through the inductor for t ≥ 0. (7)



14 For the circuit shown below, the switch K, initially at position 1 for a long time, is changed to position 2 at time $t = 0$. Using Laplace transform technique,

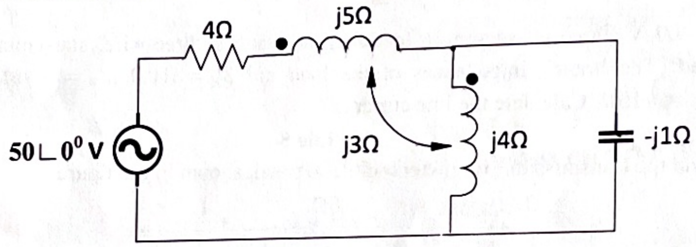
a) Find the circuit current $i(t)$ for $t > 0$. (8)

b) Obtain the expression for the voltage $V_c(t)$ across the 0.5F capacitor. (6)

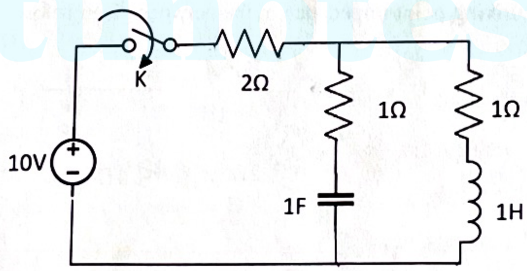


Module 3

15 a) In the circuit given below, find the current flowing through the $-j1\Omega$ capacitor. (10)



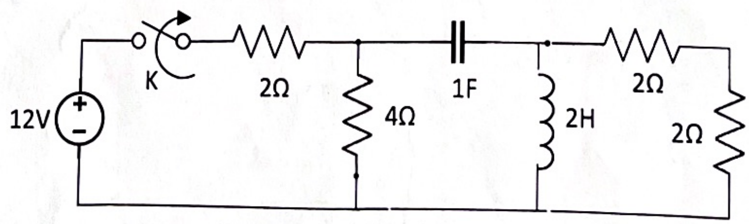
b) In the circuit given below, the switch K is closed at $t = 0$, when the initial current through the inductor is zero and initial voltage on the capacitor is 4 V. Draw the transformed circuit for $t > 0$ and write the mesh equations in s-domain. (4)



16 The switch K in the circuit given below is in closed position for a long time. At $t = 0$, the switch is opened.

a) Determine the transformed circuit for $t > 0$. (4)

b) Find the expression for the voltage across the inductor, for $t > 0$, using nodal analysis. (10)



Module 4

17 A resistor, capacitor and an inductor are connected in series with a 230 V, variable frequency AC source. When the supply frequency is varied to 50Hz, a maximum current of 2A flows and the corresponding voltage across the capacitor is 500 V. Determine,

(i) Resistance, inductance and capacitance of the circuit. (6)

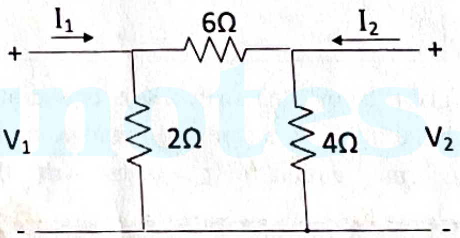
(ii) Q- factor and bandwidth of the circuit. (4)

(iii)The source frequencies at which the circuit current is $\frac{1}{\sqrt{2}}$ times the maximum current. (4)

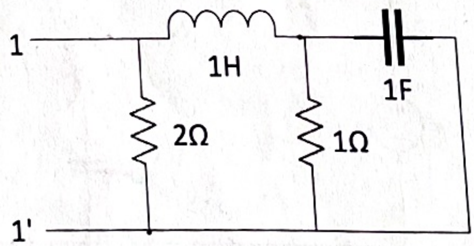
18 A 400 V, three-phase supply feeds an unbalanced three-wire, star-connected load. The branch impedances of the load are $Z_R = 10\Omega$, $Z_Y = -j5\Omega$ and $Z_B = j15\Omega$. Calculate the line currents. (14)

Module 5

19 a) Find the transmission parameters of the network shown in the figure. (8)



b) Find the driving point impedance of the network given below. (6)



20 a) Discuss the series and cascade interconnection of two port networks. (8)

b) The Y parameters of a two port network are $Y_{11} = 3\mho$, $Y_{12} = -1\mho$, $Y_{21} = -1\mho$ and $Y_{22} = 2\mho$. Determine the equivalent T-network. (6)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Scheme for Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key

Third Semester B.Tech Degree (R) Examination December 2020 (2019 scheme)

Course Code: EET201

Course Name: CIRCUITS AND NETWORKS

Max. Marks: 100 Duration: 3 Hours

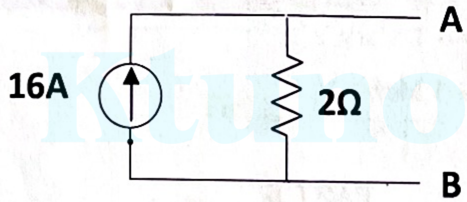
PART A

Answer all questions. Each question carries 3 marks Marks

1 Statement ------------- 1.5 Marks (3)

Explanation --------------- 1.5 Mark

2 Simplification using source transformation or any technique ------ 2 Marks (3)



---------------------- 1 Mark

3 Classification with plots showing the natural response (3)

Underdamped, critically damped and overdamped circuits ----- 3 x 1= 3 Marks

4 Differential equation (KVL) of the circuit dynamics for $t > 0$ ------- 1 Mark (3)

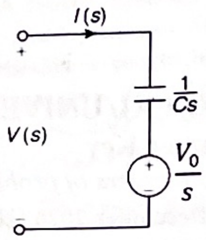
$v_c(t) = V_0 e^{-t/RC}$ ------------- 2 Marks

5 Expression for V(s) --------------- 1 Mark (3)

$v(t) = 2e^{-3t} + 2e^{-t}$ --------------- 2 Marks

6 $V(s) = \frac{1}{Cs} I(s) + \frac{V_0}{s}$ (3)

----------- 2 Marks



-------------- 1 Mark (3)

7 Explanation ---------- 2 Marks

Phasor diagram/equation showing neutral shift --------- 1 Mark (3)

8 Definition of Q factor --------- 1 Mark

Derivation -------- 2 Marks (3)

9 Network equations in terms of ABCD parameters and Z-parameters --- 1 Mark

Expression for ABCD parameters in terms of Z parameters ------- 2 Mark (3)

10 $\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$ --------- 1 Mark

$AD - BC = 1$ ---------- 1 Mark

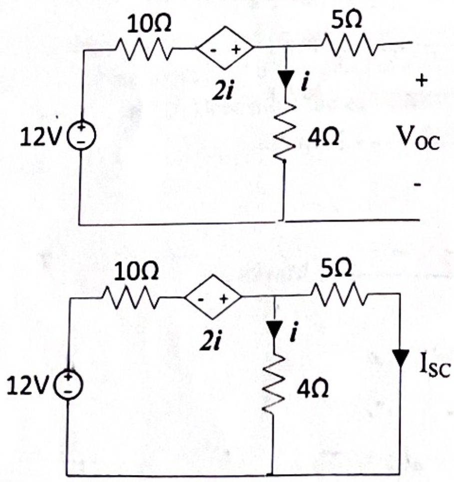
Network is reciprocal ---------1 Mark

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11 a) (14)



$V_{OC} = 4V$ ---- 3Marks $I_{SC} = 0.48A$ ----- 3 Marks

$R_{Th} = \frac{V_{OC}}{I_{SC}} = 8.33\Omega$ (any use other methods) ------ 2 Marks

Thevenin's equivalent circuit ------- 2 Marks

b) Power dissipated $= 0.3W$ ---------- 4 Marks

12 a) (14)

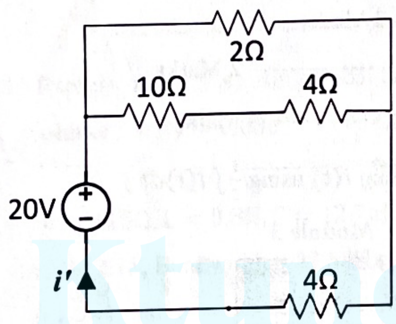


Fig. 1.

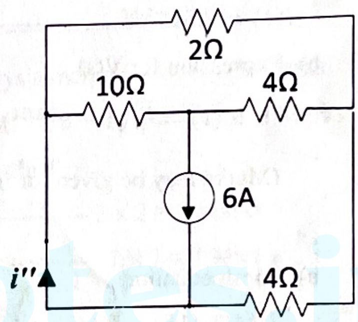


Fig. 2.

Circuit containing 20V source alone (Fig. 1.) ----------- 1.5 Marks

Current $i' = 3.48A$ ----------- 2 Marks

Circuit containing 6A source alone (Fig.2.) -------- 1.5 Marks

Current $i'' = 4.70A$ ----------- 3 Marks

(i'' can be found using delta to star transformation of resistors)

Net current $i = i' + i'' = 8.18A$ ------- 2 Marks

b) Power supplied by the source = 163.6W --------- 4 Marks

Module 2

13 a) Steady state value of inductor current = 5A ----------- 2 Marks (14)

$i(t) = 5(1 - e^{-10t/L})$ ----------- 2 Marks

$L = 14.4H$ --------------- 3 Marks

b) Circuit at $t = 0^-$ ----------- 1 Mark

Voltage across the capacitor at $t = 0^-$ is $= 3.33$V ----------- 1 Mark

Differential equation for the dynamics for $t > 0$ ----------- 1 Mark

Solution using Laplace transform:

Initial conditions : $i(0) = 0A, i'(0) = 6.66A/s$ -------- 2 Marks

$i(t) = 4.71 sin 1.414t$ ----------------- 2 Marks

14 a) Voltage across 0.25F capacitor at $t = 0^-$ is 10V -------------- 2 Mark (14)

Differential equation for $t > 0$ --------------- 2 Marks

Initial condition: $i(0^+) = 2.5A$ -------------- 1 Mark

$i(t) = 2.5e^{-1.5t}$ -------- 3 Marks

b) Expression for V(s) ------------- 2 Marks

$v_c(t) = \frac{10}{3}(1 - e^{-1.5t})$ ------------- 4 Marks

(Marks may be given for finding $i(t)$ using $\frac{1}{c}\int i(t)dt$)

Module 3

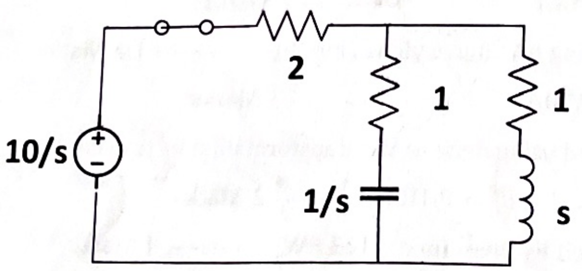
15 a) Mesh equations: (14)

$I_1(4 + j15) - I_2(j7) = 50$

$I_1(j7) + I_2(-j3) = 0$ ------------- 4 Marks

Solution: $I_2 = 27.67\angle 18.43^0 A$ -------------- 6 Marks

b) Transformed circuit



-------- 2Marks

Mesh equations:

$$\left(3 + \frac{1}{s}\right) I_1(s) - \left(1 + \frac{1}{s}\right) I_2(s) = \frac{10}{s}$$

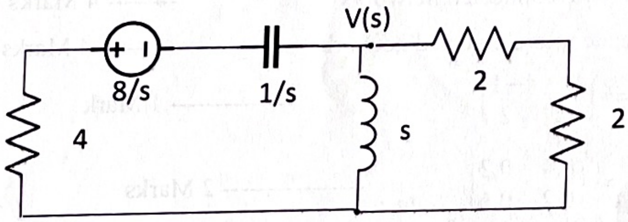
$$-\left(\frac{1}{s} + 1\right) I_1(s) + \left(s + \frac{1}{s} + 2\right) I_2(s) = 0$$

------------- 2 Marks

16 a) Initial voltage across the capacitor = 8V ----- 1Mark (14)

Initial current through the inductor = 0A --------- 1Mark

Transformed circuit ------ 2 Marks



b) Expression for V(s) using Nodal analysis method -------- 5 Marks

Solution: $v(t) = 0.68e^{-0.27t} - 4.68e^{-1.85t}$ ------------- 5 Marks

Module 4

17 (i) R = 115Ω, L = 0.8H, C = 12.7μF ---------- 3 x 2 = 6 Marks (14)

(ii) Q = 2.18, Bandwidth = 22.88Hz ----------- 2 x 2 = 4 Marks

(iii) Lower half power frequency = 38.56Hz,

Upper half power frequency = 61.44Hz ---------- 2 x 2 = 4 Marks

18 Neutral shift, $V_{NO} = 467.78\angle - 64.52^0 V$ -------------------- 4 Marks (14)

Phase voltages

$V_{RN} = 423.32\angle 85.98^0 V,\ V_{YN} = 386.93\angle 144.94^0 V,$

$V_{BN} = 698.24\angle 116.97^0 A$ ----------- 6

Marks

Phase currents

$I_{RN} = 42.33\angle 85.98^0 A,\ I_{YN} = 77.39\angle - 125.06^0 A,\ I_{BN} = 46.55\angle 27^0 A$

---------- 3 Marks

Line currents = Phase currents --------- 1 Mark

(Marks may be given for any other method of analysis such as star delta transformation method, analysis using KVL etc. Marks for steps may be awarded to students attempting the question as a balanced system)

Module 5

19 a) $A = 2.5,\quad B = 6\Omega\ ,\ C = 1.5℧,\quad D = 4$ --------- 4 x 2 = 8 Marks (14)

b) Transformed circuit ------------------ 3 marks

$$Z(s) = \frac{2(s^2 + s + 1)}{s^2 + 3s + 3}$$

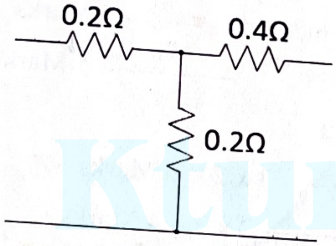
------------------- 3 marks

20 a) Series interconnected networks -------- 4 Marks

Cascade interconnected networks --------- 4 Marks

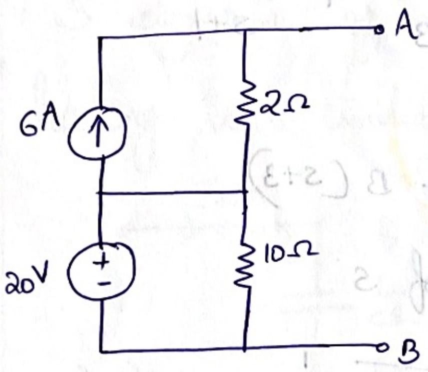
b) $[Y] = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$ -------------- 1 Mark

$[Z] = \begin{bmatrix} 0.4 & 0.2 \\ 0.2 & 0.6 \end{bmatrix}$ --------------- 2 Marks

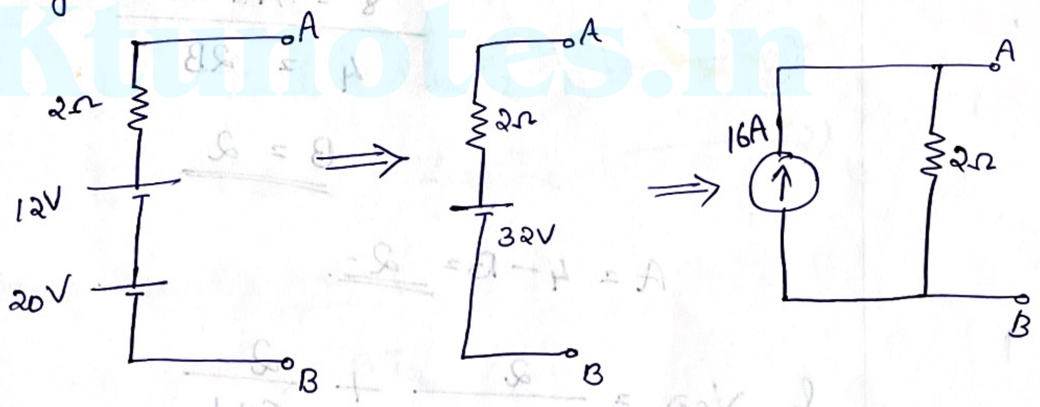


-------------------------- 3 Marks

2.



Since the resistor of 10Ω is connected in parallel with the voltage source of 20V it becomes redundant. Converting parallel combination of current source and resistor into equivalent voltage source and resistor.



5. $I(s) = \dfrac{2s+4}{s^2+4s+3}$

$V(s) = I(s) \cdot R = \dfrac{(2s+4)}{s^2+4s+3} \times 2$

$V(s) = \dfrac{4s+8}{s^2+4s+3} = \dfrac{4s+8}{(s+3)(s+1)}$

$$\frac{4s+8}{(s+3)(s+1)} = \frac{A}{s+3} + \frac{B}{s+1}$$

$$4s+8 = A(s+1) + B(s+3)$$

Equating coefficients of s

$$4 = A+B \quad \text{---} \; 1$$

Equating constants

$$8 = A+3B \quad \text{---} \; 2$$

Solving ① & ② $\Rightarrow$

$$\begin{array}{l} 4 = A+B \\ 8 = A+3B \\ \hline 4 = 2B \end{array}$$

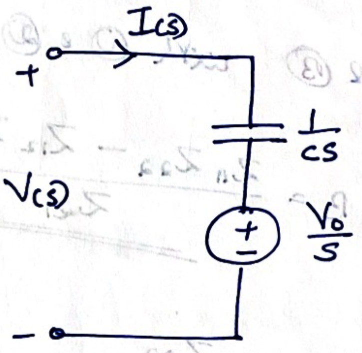
$$\underline{\underline{B = 2}}$$

$$A = 4 - B = \underline{\underline{2}}.$$

So $V(s) = \dfrac{2}{s+3} + \dfrac{2}{s+1}$

$$\underline{\underline{V(t) = 2e^{-3t} + 2e^{-t}}}.$$

6. S-domain equivalent circuit of a capacitor having an initial voltage of V_0.



$$V_{(s)} = \frac{1}{CS} I_{(s)} + \frac{V_0}{S}$$

9. Z parameter eqns

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{——— (1)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{——— (2)}$$

ABCD parameter eqns

$$V_1 = A V_2 - B I_2$$

$$I_1 = C V_2 - D I_2$$

Rewriting eqn (2) $\quad I_1 = \frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2 \quad$ ——— (A)

Substituting (A) in (1)

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - \frac{Z_{11} Z_{22}}{Z_{21}} I_2 + Z_{12} I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - \frac{(Z_{11}Z_{22} - Z_{12}Z_{21})}{Z_{21}} I_2 \quad \text{---} (3)$$

Comparing eqns (A) & (B) with (1) & (2)

$$A = \frac{Z_{11}}{Z_{21}} \; ; \qquad B = \frac{Z_{11}Z_{22} - Z_{12}Z_{21}}{Z_{21}}$$

$$C = \frac{1}{Z_{21}} \; ; \qquad D = \frac{Z_{22}}{Z_{21}}$$

10.

$$V_1 = 3V_2 - 2I_2$$

$$I_1 = 4V_2 - 3I_2$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$$

Condition for reciprocity is

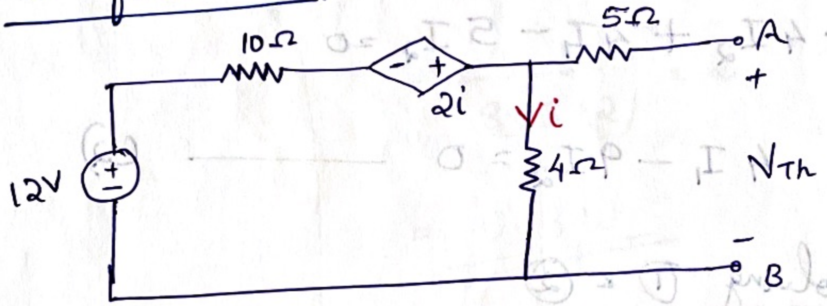
$$AD - BC = 1$$

$$3 \times 3 - 4 \times 2 = 9 - 8 = 1$$

$\therefore$ Reciprocal

11.

a) To find V_{TH}

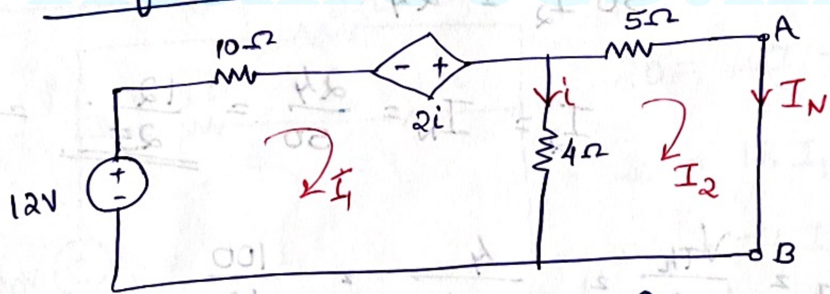


$$12 - 10i + 2i - 4i = 0$$

$$12 = 12i \; ; \quad i = 1A$$

$$V_{Th} = 4i = 4V$$

To find I_N



$$12 - 10I_1 + 2i - 4(I_1 - I_2) = 0$$

$$12 - 10I_1 + 2(I_1 - I_2) - 4(I_1 - I_2) = 0$$

$$12 - 10I_1 - 2(I_1 - I_2) = 0$$

$$12 = 12I_1 \mp 2I_2 \quad \text{———} \quad \cancel{(1)}$$

$$6 = 6I_1 \mp I_2 \quad \text{———} \quad (1)$$

$$-4(I_2 - I_1) - 5I_2 = 0$$

$$-4I_2 + 4I_1 - 5I_2 = 0$$

$$4I_1 - 9I_2 = 0 \quad \text{———} \quad (2)$$

Solving (1) & (2)

$$6I_1 \mp I_2 = 6$$

$$4I_1 - 9I_2 = 0$$

$$24I_1 \mp 4I_2 = 24$$

$$24I_1 - 54I_2 = 0$$

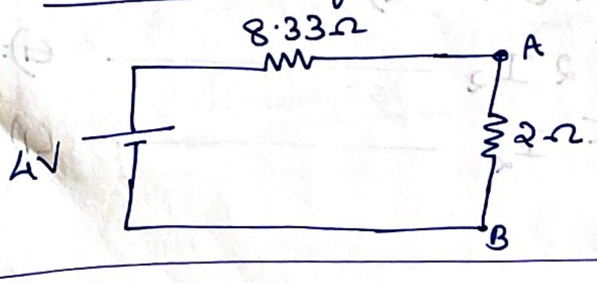
$$50I_2 = 24$$

$$I_2 = I_N = \frac{24}{50} = \frac{12}{25} = \underline{\underline{0.48A}}$$

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{4}{12/25} = \frac{100}{12}$$

$$= \frac{25}{3} = \underline{\underline{8.33\,\Omega}}$$

<u>Thevenin's equivalent Circuit</u>



11.

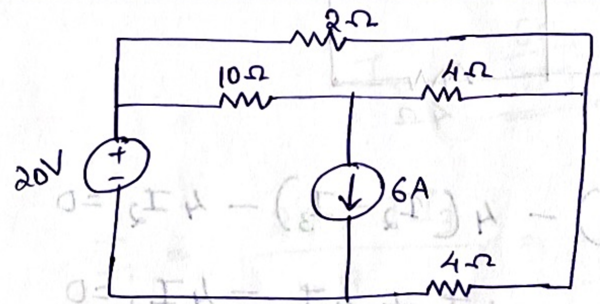
b) Power dissipated $= I^2 R$

$$= \left(\frac{4}{8.33+2}\right)^2 \times 2$$

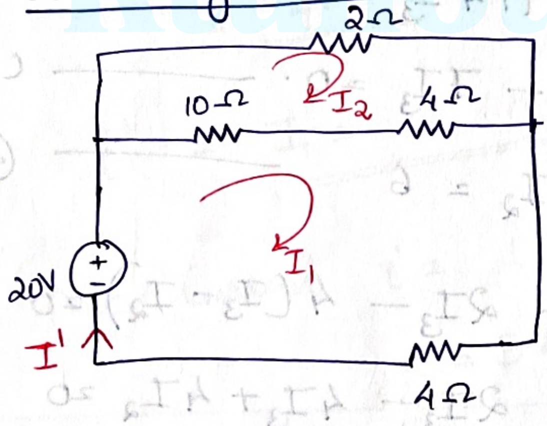
$$= \underline{\underline{0.3\,W}}$$

12.

a)



Considering 20V source alone



$$20 = 14(I_1 - I_2) + 4I_1$$

$$20 = 18I_1 - 14I_2 \quad —(1)$$

$$0 = -14I_1 + 16I_2 \quad —(2)$$

$$9I_1 - 7I_2 = 10$$

$$-7I_1 + 8I_2 = 0$$

$$63I_1 - 49I_2 = 70$$

$$-63I_1 + 72I_2 = 0$$

$$23I_2 = 70$$

$$\underline{\underline{I_2 = \frac{70}{23}}}$$

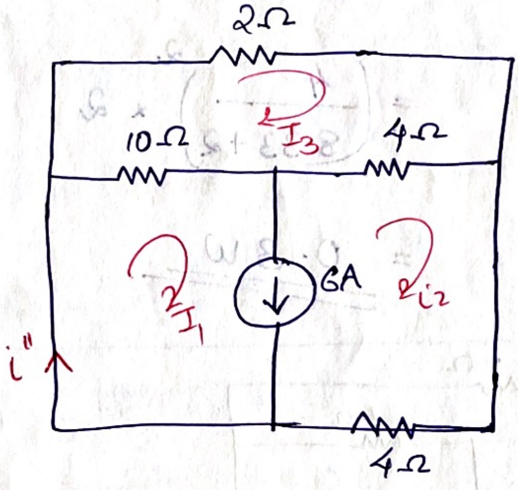
$$I_1 = \frac{10 + 7I_2}{9}$$

$$= \frac{10 + 7 \times \frac{70}{23}}{9}$$

$$= \frac{230 + 490}{207}$$

$$\underline{\underline{i' = 3.48\,A}}$$

Considering 6A Source alone



$$-10(I_1 - I_3) - 4(I_2 - I_3) - 4I_2 = 0$$

$$-10I_1 + 10I_3 - 4I_2 + 4I_3 - 4I_2 = 0$$

$$-10I_1 - 8I_2 + 14I_3 = 0$$

$$5I_1 + 4I_2 - 7I_3 = 0 \quad \text{——— (1)}$$

$$I_1 - I_2 = 6 \quad \text{——— (2)}$$

$$-10(I_3 - I_1) - 2I_3 - 4(I_3 - I_2) = 0$$

$$-10I_3 + 10I_1 - 2I_3 - 4I_3 + 4I_2 = 0$$

$$10I_1 + 4I_2 - 16I_3 = 0$$

$$5I_1 + 2I_2 - 8I_3 = 0 \quad \text{——— (3)}$$

(1) $\Rightarrow$ $5I_1 + 4(I_1 - 6) - 7I_3 = 0$

$$9I_1 - 7I_3 = 24 \quad \text{——— (A)}$$

(3) $\Rightarrow$ $5I_1 + 2I_1 - 12 - 8I_3 = 0;\quad 7I_1 - 8I_3 = 12$ ——— (B)

$$9I_1 - 7I_3 = 24$$
$$7I_1 - 8I_3 = 12$$

$$63I_1 - 49I_3 = 168$$
$$63I_1 - 72I_3 = 108$$

$$23I_3 = 60$$

$$I_3 = \frac{60}{23}$$

$$I_1 = \frac{24 + 7I_3}{9} = \frac{24 + \frac{420}{23}}{9}$$

$$= 4.7A$$

$$i'' = I_1 = 4.7A.$$

Net Current $i = i' + i''$

$$= 3.48 + 4.7 = 8.18A.$$

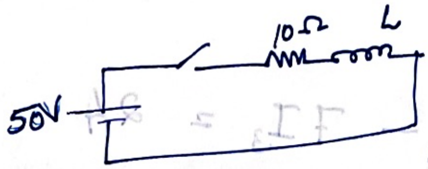
12 b) Power supplied by the 20V source

$$= 20 \times 8.18$$

$$= 163.6\ W$$

13

a. 50 V supply



$R = 10\Omega$

Steady state value of inductor current

$$I_0 = \frac{V}{R} = \frac{50}{10} = 5A$$

$$i_{(t)} = I_0\left(1 - e^{-\frac{R}{L}t}\right)$$

when $t = 1$ sec, $i_{(t)} = \frac{5}{2} = 2.5A$.

$$\frac{5}{2} = 5\left(1 - e^{\frac{-10}{L}}\right)$$

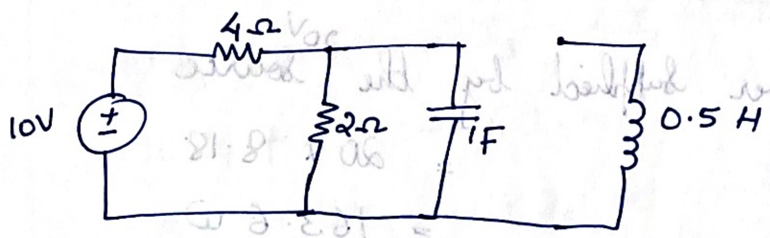
$$\frac{1}{2} = 1 - e^{\frac{-10}{L}}$$

$$e^{\frac{-10}{L}} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\frac{-10}{L} = \ln\left(\frac{1}{2}\right)$$

$$L = \frac{-10}{\ln\left(\frac{1}{2}\right)} = 14.4\ H$$

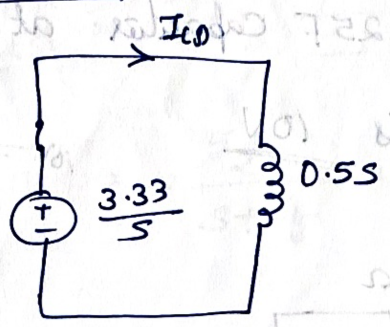
b. Circuit at $t = 0^-$



Voltage across the capacitor at $t = 0^-$ is $\frac{10}{(4+2)} \times 2$

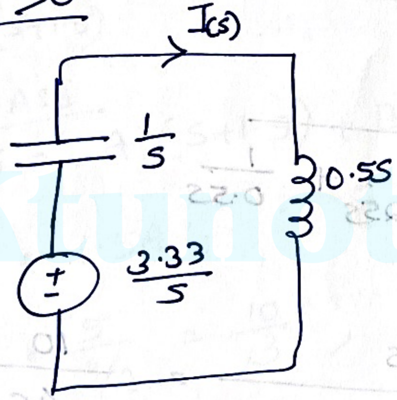
$$= \frac{10 \times 2}{6} = 3.33V$$

For $t=0$



$$I(s) = \frac{\frac{3.33}{s}}{0.5s} = \frac{3.33/s}{\frac{1}{2}s} = \frac{6.66}{s^2}$$

$t>0$



$$I(s) = \frac{3.33/s}{\frac{1}{s}+\frac{1}{2}s} = \cancel{\frac{3.33/s}{}} = \frac{3.33}{1+\frac{1}{2}s^2}$$

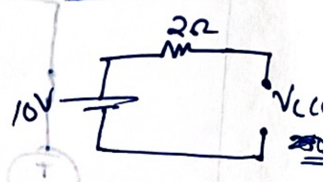
$$I(s) = \frac{3.33}{\frac{2+s^2}{2}} = \frac{6.66}{s^2+2}$$

$$I(s) = \frac{6.66}{s^2+(\sqrt{2})^2} = 4.71 \times \frac{\sqrt{2}}{s^2+(\sqrt{2})^2}$$

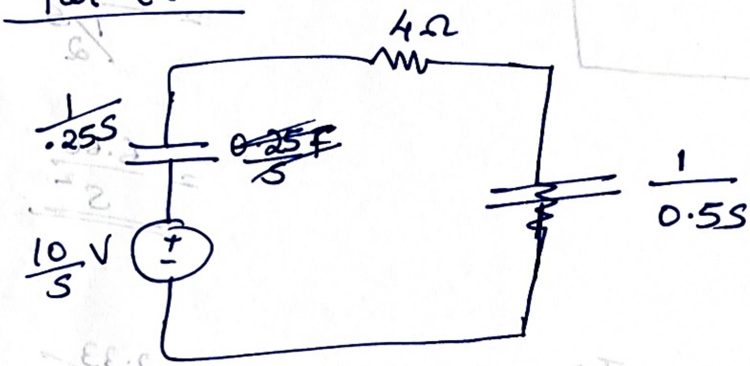
So $i(t) = 4.71 \sin 1.414t$.

14 a) Voltage across 0.25F Capacitor at $t = 0^-$

is ~~10V~~



For $t > 0$



$$I(s) = \frac{10/s}{4 + \frac{1}{0.25s} + \frac{1}{0.5s}}$$

$$= \frac{10/s}{4 + \frac{4}{s} + \frac{2}{s}} = \frac{10}{4s+6} = \frac{5}{2s+3}$$

$$I(s) = \frac{5}{2\left(s+\frac{3}{2}\right)}$$

$$\cancel{\frac{5/2}{s+\frac{3}{2}} = A}$$

$$i(t) = 2.5\, e^{-1.5t}$$

14.

b.

$$V_c(s) = I(s) \cdot \frac{1}{Cs}$$

$$= \frac{5/2}{s+\frac{3}{2}} \times \frac{1}{0.5s}$$

$$= \frac{5/2}{s+1.5} \times \frac{2}{s} = \frac{5}{s(s+1.5)}$$

$$\frac{5}{s(s+1.5)} = \frac{A}{s} + \frac{B}{s+1.5}$$

Find A&B

$$5 = A(s+1.5) + Bs$$

$$0 = A + B$$

$$5 = 1.5A \; ;$$

$$B = \frac{-10}{3}$$

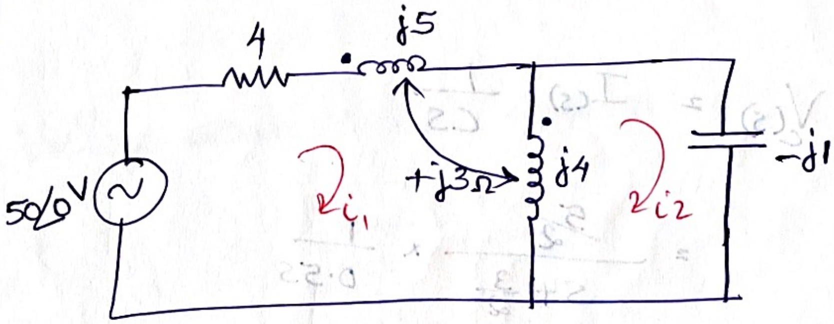
$$A = \frac{5}{3/2} = \frac{10}{3} \; ;$$

$$V_c(s) = \frac{10/3}{s} + \frac{-10/3}{s+1.5}$$

$$V_c(t) = \frac{10}{3} \times 1 - \frac{10}{3} e^{-1.5t}$$

$$V_c(t) = \frac{10}{3}\left[1 - e^{-1.5t}\right]$$

15 a)



Mesh I.

$$50\angle 0 - 4i_1 - j5i_1 - j3(i_1 - i_2) - j4(i_1 - i_2) - j3\,i_1 = 0$$

$$50\angle 0 = 4i_1 + j5i_1 + j3i_1 \mp j3i_2 + j4i_1 \mp j4i_2 + j3i_1$$

$$50\angle 0 = (4 + j15)\,i_1 - j7\,i_2 \quad \text{——— (1)}$$

Mesh 2

$$-j4(i_2 - i_1) + j3i_1 + j1i_2 = 0$$

$$-j4i_2 + j4i_1 + j3i_1 + j1i_2 = 0$$

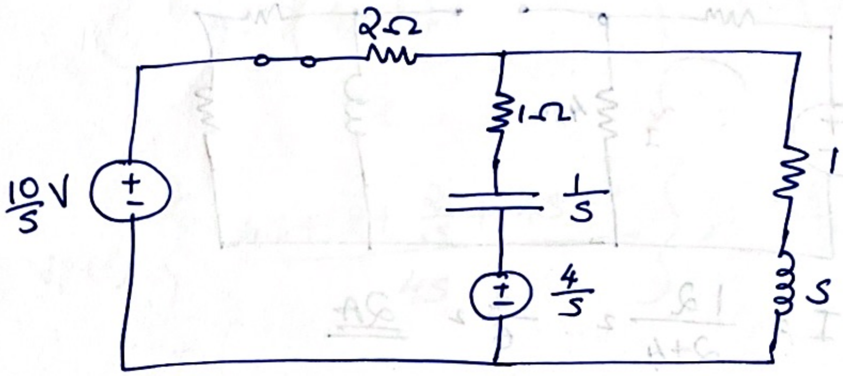
$$j7i_1 - j3i_2 = 0 \quad \text{——— (2)}$$

Solve eqns (1) & (2)

$$I_2 = 27.67\angle 18.43^\circ \text{ A}$$

15. b)

Transformed Circuit



Mesh 1

$$\left(3+\frac{1}{s}\right) I_1(s) - \left(1+\frac{1}{s}\right) I_2(s) = \frac{6}{s} \quad \text{--- (1)}$$

Mesh 2

$$-\left(\frac{1}{s}+1\right) I_1(s) + \left(s+\frac{1}{s}+2\right) I_2(s) = \frac{4}{s} \quad \text{--- (2)}$$

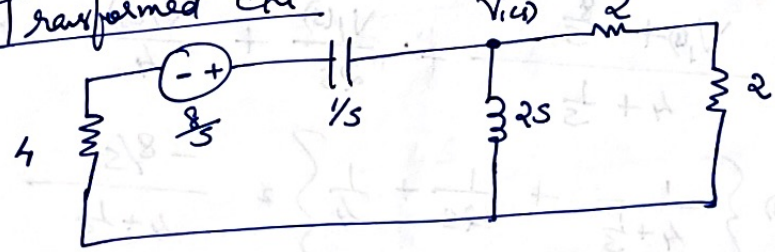
16. a)

Initial voltage across Capacitor

$$= \frac{12}{2+4} \times 4 = \frac{12}{6} \times 4 = \underline{\underline{8\ V}}$$

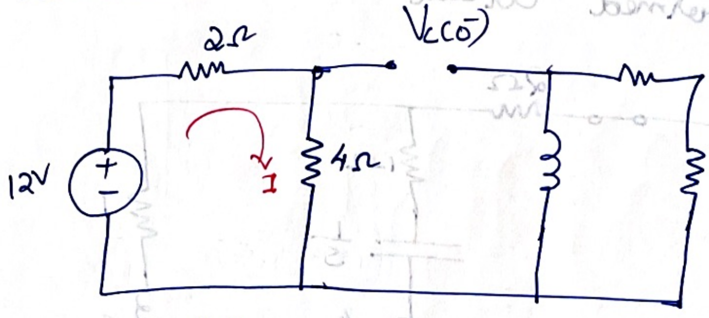
Initial Current through inductor $= \underline{\underline{0\ A}}$

Transformed ckt



16 b

At $t = 0^-$



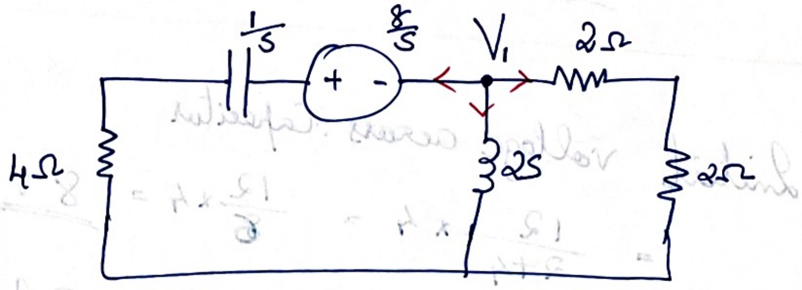
$$I = \frac{12}{2+4} = \frac{12}{6} = \underline{2A}$$

$V_c(0^-)$ = Voltage across 4Ω resistor

$$= 4I = 4 \times 2 = \underline{8V}$$

At $t = 0^+$

S/W is opened.



Applying KCL to node 1

$$\frac{V_1(s) + \frac{8}{s}}{4 + \frac{1}{s}} + \frac{V_1(s)}{2s} + \frac{V_1(s)}{4} = 0$$

$$V_1(s) \left\{ \frac{1}{4+\frac{1}{s}} + \frac{1}{2s} + \frac{1}{4} \right\} = \frac{-8/s}{4+\frac{1}{s}}$$

$$V_1(s)\left\{\frac{4s+2\left(4+\frac{1}{s}\right)+s\left(4+\frac{1}{s}\right)}{4s\cancel{\left(4+\frac{1}{s}\right)}}\right\}=\frac{-8/s}{\cancel{\left(4+1/s\right)}}$$

$$V_1(s)\left\{\frac{4s+8+\frac{2}{s}+4s+1}{4s}\right\}=\frac{-8}{s}$$

$$V_1(s)\left\{\frac{8s+9+\frac{2}{s}}{4s}\right\}=\frac{-8}{s}$$

$$\cancel{V_1(s)\left\{2s+\frac{9}{4}+\frac{1}{2s}\right\}=-8}$$

$$V_1(s)\left\{\frac{8s^2+9s+2}{\cancel{4}s}\right\}=-8$$

$$V_1(s)=\frac{-8\times 4s}{8s^2+9s+2}$$

$$=\frac{-4s}{s^2+\frac{9}{8}s+\frac{1}{4}}=\frac{-4s}{s^2+1.125s+0.25}$$

$$=\frac{-4s}{(s+0.305)(s+0.82)}$$

$$V_1(s) = \frac{-4s}{(s+0.305)(s+0.82)} = \frac{A}{s+0.305} + \frac{B}{s+0.82}$$

X'ing with $(s+0.305)(s+0.82)$

$$-4s = A(s+0.82) + B(s+0.305)$$

Comparing Coeff. of s

$$-4 = A+B; \quad B = -4-A$$

Comparing Constants

$$0 = 0.82A + 0.305B$$

$$0 = 0.82A + 0.305(-4-A)$$

$$0 = 0.82A - 1.22 - 0.305A$$

$$0 = 0.515A - 1.22.$$

$$A = \frac{1.22}{0.515} = \underline{\underline{2.37}}.$$

$$B = -4-A = -4-2.37 = \underline{\underline{-6.37}}$$

So $$V_1(s) = \frac{2.37}{s+0.305} - \frac{6.37}{s+0.82}$$

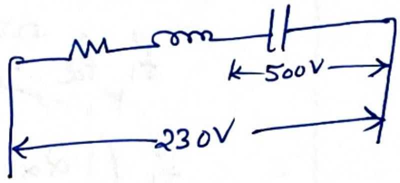
Taking Inverse L.T on both sides

$$V(t) = \underline{\underline{2.37\,e^{-0.305t} - 6.37\,e^{-0.82t}}}$$

17. a

Resonance

i) $R = \frac{V}{I} = \frac{230}{2} = 115\,\Omega$



$V_c = 500V$

$IX_c = 500$

$X_c = \frac{500}{2}$

$1/C\omega = \frac{500}{2}$; $C\omega = \frac{2}{500}$

$C = \frac{2}{2\times\pi\times 50\times 500}$

$= 12.7\,\mu F$

$X_L = X_c$

$L\omega = \frac{1}{C\omega}$

$L = \frac{1}{C\omega^2} = \frac{1}{12.7\times10^{-6}\times(2\pi f)^2}$

$= 0.8\ H$

ii) Q Factor $= \frac{1}{R}\sqrt{\frac{L}{C}}$ or $\frac{1}{\omega_0 CR}$

$= \frac{1}{115}\sqrt{\frac{0.8}{12\times10^{-6}}} = 2.18$

$B.W = \frac{f_0}{Q_0} = \frac{50}{2.18} = 22.88\ Hz$

iii) $f_2 - f_1 = 22.88 \implies f_2 = 22.88 + f_1$

$\omega_1 \omega_2 = \omega_0^2$

$f_1 f_2 = f_0^2$

$f_1 (22.88 + f_1) = 50^2$

$f_1^2 + 22.88 f_1 - 2500 = 0$

$$f_1 = \frac{-22.88 \pm \sqrt{(22.88)^2 + 4 \times 2500}}{2}$$

$$= \frac{-22.88 \pm 102.58}{2}$$

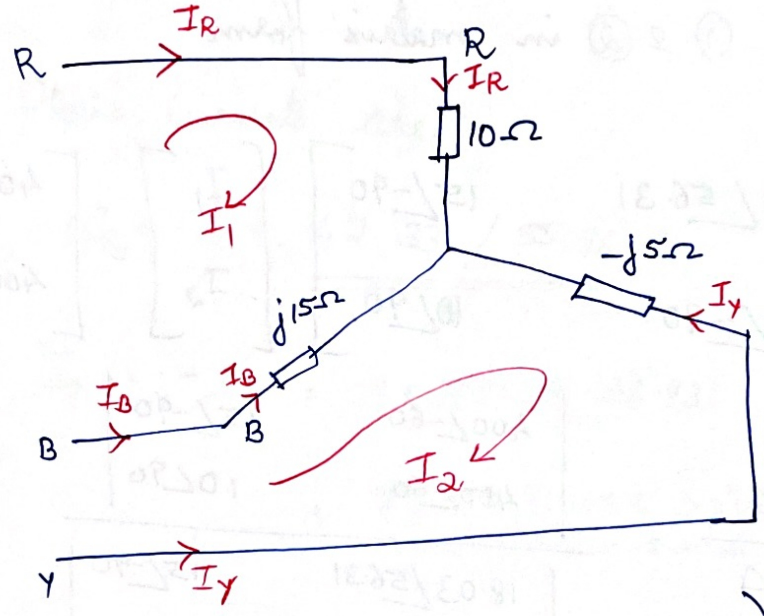
$= \underline{\underline{39.85 \text{ Hz}}}$

$f_2 = BW + f_1$

$= 22.88 + 39.85$

$= \underline{\underline{62.73 \text{ Hz}}}$.

18.



$V_{RY} = 400\angle 0$

$V_{YB} = 400\angle -120$

$V_{BR} = 400\angle -240$

KVL to loop 1

$$V_{RB} = 10 I_1 + j15(I_1 - I_2)$$

$$-1 \times V_{BR} = (10 + j15) I_1 - j15 I_2$$

$$-1 \times 400\angle -240 = (10 + j15) I_1 - j15 I_2$$

$$400\angle -60 = 18.03\angle 56.31 + 15\angle -90 \; I_2 \quad \text{——} (1)$$

KVL to loop 2

$$V_{BY} = j15(I_2 - I_1) - j5 I_2$$

$$-1 \times V_{YB} = j15 I_2 - j15 I_1 - j5 I_2$$

$$-1 \times 400\angle -120 = -j15 I_1 + j10 I_2$$

$$400\angle 60 = 15\angle -90 \; I_1 + 10\angle 90 \; I_2 \quad \text{——} (2)$$

Equations ① & ② in matrix form

$$\begin{bmatrix} 18.03\angle 56.31 & 15\angle -90 \\ 15\angle -90 & 10\angle 90 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 400\angle -60 \\ 400\angle 60 \end{bmatrix}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} 400\angle -60 & 15\angle -90 \\ 400\angle 60 & 10\angle 90 \end{vmatrix}}{\begin{vmatrix} 18.03\angle 56.31 & 15\angle -90 \\ 15\angle -90 & 10\angle 90 \end{vmatrix}}$$

$$= \frac{5291.5\angle 109.1}{125\angle 53.14} = \underline{\underline{42.33\angle 55.96^\circ \text{ A}}}$$

$$I_2 = \frac{\Delta_2}{\Delta}$$

$$= \frac{\begin{vmatrix} 18.03\angle 56.31 & 400\angle -60 \\ 15\angle -90 & 400\angle 60 \end{vmatrix}}{\begin{vmatrix} 18.03\angle 56.31 & 15\angle -90 \\ 15\angle -90 & 10\angle 90 \end{vmatrix}} = \frac{9673.82\angle 78.07}{125\angle 53.14}$$

$$= \underline{\underline{77.39\angle 24.93^\circ \text{ A}}}.$$

Phase currents are

$$I_R = I_1 = \underline{\underline{42.33 \angle 55.96^\circ \text{ A}}}$$

$$I_Y = -I_2 = -1 \times 77.39 \angle 24.93$$

$$= \underline{\underline{77.39 \angle -155.07^\circ \text{ A}}}$$

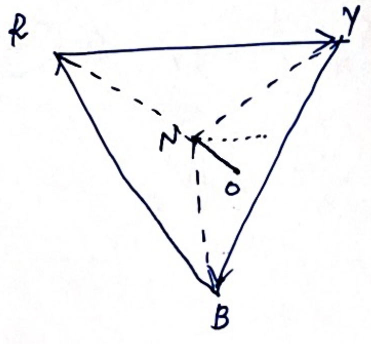
$$I_B = I_2 - I_1 = 77.39 \angle 24.93 - 42.33 \angle 55.96$$

$$= \underline{\underline{46.55 \angle -3.02^\circ}} \text{ A}$$

Line Currents = Phase Currents $\therefore$ Star Connection

Neutral shift voltage $V_{NO} = V_{RN} - V_{RO}$

$$= \frac{400}{\sqrt{3}} \angle -30 - I_R Z_R$$



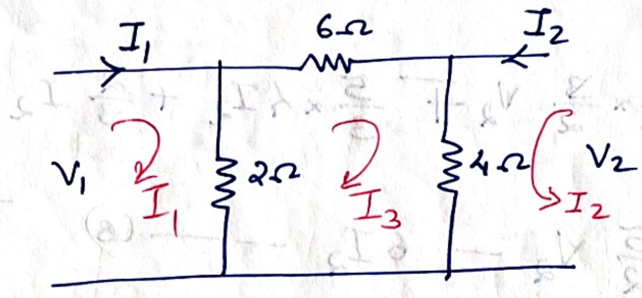
$$= 230.94 \angle -30 - 42.33 \angle 55.96 \times 10$$

$$= 230.94 \angle -30 - 423.3 \angle 55.96$$

$$= \underline{\underline{467.7 \angle -94.53^\circ}} \text{ V}$$

19.

a)



Mesh 1 $\rightarrow$ $V_1 = 2(I_1 - I_3)$ ——— 1

Mesh 2 $\rightarrow$ $V_2 = 4(I_2 + I_3)$ ——— 2

Mesh 3 $\rightarrow$ $2(I_3 - I_1) + 6I_3 + 4(I_3 + I_2) = 0$

$-2I_1 + 4I_2 + 12I_3 = 0$ ——— 3

$I_3 = \frac{1}{6} I_1 - \frac{1}{3} I_2$ ——— 4

$2 - \frac{1}{3}$

(4) in (1)

$V_1 = 2I_1 - \frac{2}{6} I_1 + \frac{2}{3} I_2$

$V_1 = \frac{5}{3} I_1 + \frac{2}{3} I_2$ ——— (5)

$4 - \frac{4}{3}$

(2) $\Rightarrow$ $V_2 = 4I_2 + \frac{4}{6} I_1 - \frac{4}{3} I_2$

$\frac{8}{3} \times \frac{3}{2}$

$V_2 = \frac{2}{3} I_1 + \frac{8}{3} I_2$

$I_1 = \frac{3}{2} V_2 - 4 I_2$ ——— (A)

$C = \frac{3}{2}$; $D = 4$

⑤ $\Rightarrow$

$$V_1 = \frac{5}{3} \times \frac{3}{2} V_2 \mp \frac{5}{3} \times 4 I_2 + \frac{2}{3} I_2$$

$-\frac{20}{3} + \frac{2}{3}$

$-\frac{18}{3}$

$$V_1 = \frac{5}{2} V_2 - 6 I_2 \quad \text{——— (B)}$$

$A = \frac{5}{2}$ $\quad C = \frac{3}{2}$

$B = 6$ $\quad D = 4$

6. $\left\{\left\{\frac{1}{s} \| 1\right\} + s\right\} \| 2$

$$\frac{\left\{\frac{\frac{1}{s} \times 1}{\frac{1}{s} + 1} + s\right\} \times 2}{\frac{\frac{1}{s} \times 1}{\frac{1}{s} + 1} + s + 2} = \frac{\left(\frac{1}{s+1} + s\right) 2}{\frac{1}{s+1} + s + 2}$$

$$= \frac{(s^2 + s + 1)\, 2}{s^2 + 3s + 2 + 1} = \frac{2(s^2 + s + 1)}{s^2 + 3s + 3}$$

20 b.

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}^{-1} = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$[Z] = \begin{bmatrix} 0.4 & 0.2 \\ 0.2 & 0.6 \end{bmatrix}$$